

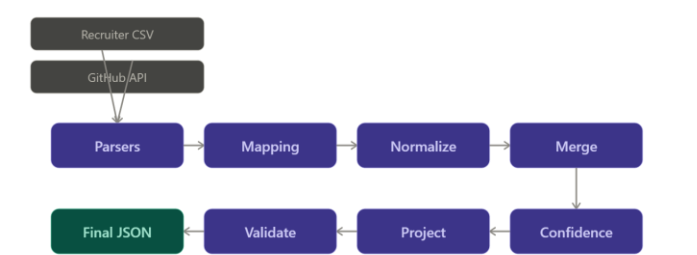
MULTI-SOURCE CANDIDATE DATA TRANSFORMER

Technical Design

1. Design Philosophy

A deterministic, explainable transformation engine that converts heterogeneous candidate data into a single canonical profile while preserving provenance and confidence. The design prioritizes modularity, configurability, and robustness by separating source-specific parsing from the core transformation engine.

2. System Architecture



3. Canonical Schema

Canonical Model

A typed Candidate dataclass serves as the single internal representation. Nested structures (location, links, skills) remain dictionaries/lists to support flexible path-based projection while preserving schema consistency. Data is normalized to E.164 phone numbers, ISO-3166 country codes, and canonical skill names.

Fields

candidate_id • full_name • emails[] • phones[]
location{} • links{} • headline • years_experience
skills[] • experience[] • education[] • provenance{} • overall_confidence

Sources Implemented

Structured: Recruiter CSV

Unstructured: GitHub Public API (free-text bio, inferred skills from repository languages, and source-specific fields requiring canonical mapping).

4. Merge / Conflict-Resolution Policy

Field	Strategy	Rationale
full_name, headline, years_experience	Priority: CSV → GitHub	CSV preferred
emails, phones	Normalize → Union → Deduplicate	Preserve all contacts
location	First non-null (CSV priority order)	Deterministic
links	Merge by type	Independent link types

skills	Union + Confidence-Weighting	Multi-source corroboration
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Merge decisions are deterministic. Every selected value is accompanied by field-level provenance {field, value, source, method}.

5. Confidence Strategy

Rule-based (no ML) for complete explainability. Each populated field contributes a fixed weight (e.g. full_name = **0.20**, emails = **0.18**, phones = **0.10**, skills = **0.10**), while supplementary fields (location, links, experience, education, etc.) contribute proportionally smaller weights. An additional **+0.03** bonus is awarded when a field is corroborated by multiple sources, and a **-0.15** penalty is applied for each missing required field (full_name, emails). Scores are clamped to **[0,1]** and returned with human-readable reasons for explainability.

6. Runtime Configuration

Runtime-Configurable Output:

- Field selection
- Field remapping (from)
- Per-field normalization
- Provenance / confidence toggles
- Missing-value policy (null, omit, error)
- Nested path support (links.github, skills[].name)

The transformation engine remains unchanged. **Only the configuration determines the output schema.**

7. Edge Cases

Scenario	Behaviour
Missing GitHub	Continue with CSV
API Failure	Log and continue
Invalid Email	Flag record
Conflicting values	Deterministic merge

8. Engineering Trade-offs

- Chose **GitHub over resume parsing** to focus on the transformation engine rather than **OCR/layout extraction**.
- Used one **typed canonical Candidate model** while keeping nested structures dictionary-based for **flexible projection**.
- Implemented a **CLI instead of a GUI**, aligning with the assignment's emphasis on **engine correctness** over interface polish.